

DIGITAL DIARY

PYTHON PROJECT | SESSION 1

Building Your Own Contact Book

The Lost Number Problem

Imagine you forgot your friend Aarav's phone number and need to call him for homework help! How do we keep all important numbers safe and easy to find?

What is a Digital Diary?

A Digital Diary is like a modern address book on your computer!

It uses a special Python tool called a **Dictionary** to store information as **Keys** and **Values**.

Keys and Values

Think of it this way: **Key** = **Label**, and **Value** = **Information Stored**!

Indian Examples:

- **Kirana store ledger:** Customer name (Key) → Amount owed (Value)
- **Aadhaar card:** Unique number (Key) → Your details (Value)
- **Mobile contacts:** 'Mummy' (Key) → Phone number (Value)

Python Dictionaries Refresher

You already know how to use these!

```
contacts = {  
    'Daddy': '9876543210',  
    'Mummy': '9123456789',  
    'Diya': '9988776655'  
}
```

Let's Practice!

Here are some contacts:

- **Aarav:** 9876543210
- **Saanvi:** 9123456789
- **Arjun:** 9988776655

Question: Can you tell the phone number for Saanvi?

Starting Our Project

Let's begin with an empty dictionary:

```
contacts = {}
```

Our Menu Options:

1. Add Contact
2. View All Contacts
3. Search Contact
4. Exit

Adding Contacts

Here is how we add a new contact to our dictionary step by step!

```
def add_contact():  
    name = input("Enter name: ")  
    phone = input("Enter phone number: ")  
    contacts[name] = phone  
    print("Shabaash! Contact added.")
```

The Science: The name becomes the key, and the phone number becomes the value!

Viewing All Contacts

The 'for loop' goes through each contact one by one and prints the name with its phone number!

```
def view_contacts():  
    for name, phone in contacts.items():  
        print(f"{name}: {phone}")
```

Searching Contacts

The Truecaller Logic: Just like Truecaller helps you find who's calling, our search feature finds contacts instantly!

```
def search_contact():  
    name = input("Enter name: ")  
    if name in contacts:  
        print(f"Number: {contacts[name]}")  
    else:  
        print("Contact not found.")
```

Our Menu Loop

Think of a while loop as a door that stays open until you decide to leave! Our menu keeps showing options until you choose Exit.

```
while True:
    print("1. Add 2. View 3. Search 4. Exit")
    choice = input("Pick: ")
    if choice == '4':
        break
```

Recap: What We Built

- Used dictionaries to store contacts.
- Keys = Names, Values = Phone numbers.
- Built Add, View, and Search functions.
- Created a UI loop to keep the app running.

Shabaash!

You built your first digital diary!

Now It's Your Turn!

Start coding your contact book and make it yours. Be creative, explore new ideas, and don't be afraid to try different things.

You have all the skills you need!

Thank you for coding with us today!

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